

CSC311 Introduction to Machine Learning Week 3 Exercises

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

Training

W3-T1. Compute the mean square error (the average square loss across a data set) of the linear regression model (hypothesis) $y = 0.5x + 1$ on the following training data set

$$\{(x^{(1)} = 3, t^{(1)} = 2), (x^{(2)} = 2, t^{(2)} = 3), (x^{(3)} = 1, t^{(3)} = 0)\}$$

Is this cost smaller or larger than the model $y = 1x + 0.5$?

W3-T2. Use the direct solution to find the choice of w and b that would minimize the mean square error cost for the data set

$$\{(x^{(1)} = 3, t^{(1)} = 2), (x^{(2)} = 2, t^{(2)} = 3), (x^{(3)} = 1, t^{(3)} = 0)\}$$

You should be able to obtain a system of 2 equations with 2 unknowns by setting $\frac{\partial \mathcal{E}}{\partial w} = 0$ and $\frac{\partial \mathcal{E}}{\partial b} = 0$. Explain why your choice of w and b a minima of the MSE cost function (e.g., rather than a maxima or saddle point).

W3-T3. Describe how the linear regression model $y = \mathbf{w}^T \mathbf{x} + b$ can be expressed in the form $y = \tilde{\mathbf{w}}^T \tilde{\mathbf{x}}$ (i.e. b is no longer explicitly present). What changes need to be made to the original vectors $\mathbf{w}$ and $\mathbf{x}$? What happens to their shapes?

W3-T4. Suppose that we would like to train a linear model $\mathbf{y} = \mathbf{w}^T \mathbf{x}$ on the following data set using gradient descent. We will use the square loss, and thus the mean square error cost function.

$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 2 \end{bmatrix} \quad \mathbf{t} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

- (a) Suppose we initialize our weights to $\mathbf{w} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. What is the initial value of the cost function?
- (b) Compute the value of the gradient $\nabla_{\mathbf{w}} \mathcal{E}$ at $\mathbf{w} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.
- (c) Suppose that we set the learning rate $\alpha = 0.2$. What is the new value of $\mathbf{w}$ after one iteration of gradient descent?
- (d) What is the new value of the cost function after one iteration of gradient descent?
- (e) Use the linear regression model (the hypothesis) from part (c) to make a prediction for the data point $\mathbf{x} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$.
- (f) During our lecture, we visualized the linear regression cost function in weight space and saw that the contours were ellipses. For this question, write the cost in the form $\mathcal{E}(w_1, w_2) = c_1(w_1 - d_1)^2 + c_2(w_2 - d_2)^2 + \mathcal{E}_0$, then use a graphing software to sketch the contour $\mathcal{E}(w_1, w_2) = 1$ (or do so by hand).

W3-T5. Suppose you are training a linear regression model using gradient descent optimization. What can happen if the learning rate α is too small? What about too large?

Validation

W3-V1. One of the core ideas in machine learning is the idea of transforming *learning problems* into *optimization problems*. How is this idea used in learning decision trees? What about in linear regression?

W3-V2. Suppose that we are training a linear regression model $y = \mathbf{w}^T \mathbf{x}$, but we would like to use the following loss function (called the *absolute loss*):

$$\mathcal{L}_1(y, t) = |y - t|$$

Compute the gradients $\nabla_{\mathbf{w}} \mathcal{L}_1$. Start by computing the individual derivatives $\frac{\partial \mathcal{L}_1}{\partial w_j}$, and then vectorize the result.

W3-V3. Consider a linear regression problem on a data set in which each data point $(\mathbf{x}^{(i)}, t^{(i)})$ is associated with a weighting factor $r^{(i)} > 0$, so that the cost function becomes:

$$\mathcal{E}_w(\mathbf{w}) = \frac{1}{2N} \sum_i r^{(i)} (\mathbf{w}^T \mathbf{x}^{(i)} - t^{(i)})^2.$$

(a) Showing all your work, show that

$$\frac{\partial \mathcal{E}_w}{\partial w_j} = \frac{1}{N} \sum_{i=1}^N r^{(i)} x_j^{(i)} (\mathbf{w}^T \mathbf{x}^{(i)} - t^{(i)}).$$

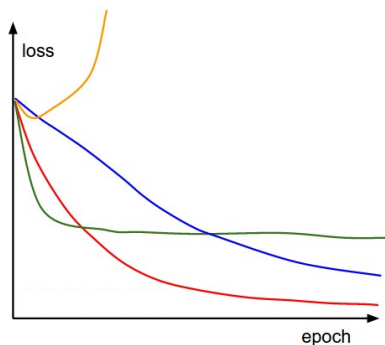
(b) Vectorize the above equation to show that

$$\nabla_{\mathbf{w}} \mathcal{E}_w = \frac{1}{N} \mathbf{X}^T \text{diag}(\mathbf{r})(\mathbf{X}\mathbf{w} - \mathbf{t}),$$

where $\mathbf{X}$ is the $N \times D$ data matrix, $\mathbf{t}$ is the $N \times 1$ vector of targets, $\mathbf{r}$ is the $N \times 1$ vector of weighting factors, and $\text{diag}(\mathbf{r})$ is the $N \times N$ diagonal matrix where the diagonal entries are those of $\mathbf{r}$.

(c) Find the direct solution to $\nabla_{\mathbf{w}} \mathcal{E}_w = \frac{1}{N} \mathbf{X}^T \text{diag}(\mathbf{r})(\mathbf{X}\mathbf{w} - \mathbf{t}) = 0$ for the optimal $\mathbf{w}$. You may assume that $\mathbf{X}^T \text{diag}(\mathbf{r}) \mathbf{X}$ is invertible. You should get the answer $\mathbf{w} = (\mathbf{X}^T \text{diag}(\mathbf{r}) \mathbf{X})^{-1} \mathbf{X}^T \text{diag}(\mathbf{r}) \mathbf{t}$.

W3-V4. The below figure shows four training curves resulting from learning linear regression models using gradient descent. The training curves each show the training cost (labeled “loss”) over each epoch. The training curves differ in the learning rate used to learn the linear regression model. Which of the four curves correspond to a learning rate that is (a) very high, (b) high, (c) good, and (d) low? Label the four curves.



(Note: Here, an *epoch* is a pass through the training set. We will explore the difference between an *epoch* and an *iteration* in future lectures. For now, they measure the same thing.)

Generalization

- W3-G1. Prove that the second derivative of the mean square error with respect to an arbitrary weight is non-negative. (i.e. $\frac{\partial^2 \mathcal{E}}{\partial w_j^2} \geq 0$) (This is a necessary condition for the mean square error to be convex for linear regression, but it is not a full proof.)
- W3-G2. Suppose we are trying to decide whether to use the square loss $\mathcal{L}_{SE}(y, t) = (y - t)^2$ or the absolute loss $\mathcal{L}_{AE}(y, t) = |y - t|$ in order to train a linear regression model. An outlier is a data point with either an input or output feature that is far outside the normal range of that feature. Which of the two losses would make our model *more sensitive to outliers*? Explain why (intuitively). (Hint: You may treat the absolute value as a piecewise function.)
- W3-G3. For this question, we will make some assumptions about the (probability) distribution of our data set in order to understand why square loss is often used in linear regression models (beyond just mathematical convenience). In particular, we will assume that

$$t = f(\mathbf{x}) + \epsilon,$$

where f is some deterministic mapping from the input to the output space, and ϵ is a random variable with a Gaussian distribution. In particular, ϵ has mean 0 and variance σ^2 . Here, we are treating $\mathbf{x}$ also as a random variable, and its distribution is called the *data distribution*.

- (a) Recall that the probability distribution function of such a Gaussian is:

$$p(\epsilon = e) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{e^2}{2\sigma^2}\right)$$

Show that the expected value of such a Gaussian is 0.

- (b) Show that the expected value of the random variable t is $f(\mathbf{x})$.
- (c) In the next two steps, we will show that if we minimize the a square loss function to obtain the optimal prediction y for a new value of $\mathbf{x}$, then we must have $y = f(\mathbf{x})$. To do so, let us set up the notation for this problem. We treat the input features in $\mathbf{x}$ as known values, and we will treat $t = f(\mathbf{x}) + \epsilon$ as a random variable (with ϵ coming from a normal distribution as above. We wish find y to minimize $L = \frac{1}{2}(y - t)^2$ (which we can also treat as a random variable). Here, we will assume that f is deterministic and known. Now, show that if we take $y = f(\mathbf{x})$, that $\mathbb{E}[L] = \frac{1}{2}\sigma^2$
- (d) Show that if we take y to be any value, say $y = f(\mathbf{x}) + \hat{y}$ with $\hat{y} > 0$, $\hat{y} \in \mathbb{R}$, that $\mathbb{E}[L] > \frac{1}{2}\sigma^2$.